

Protection mathematics

T04 — solid-state protection for a 120 VDC lunar microgrid. Symbols used throughout protection.py.

Symbols

V	feeder nominal voltage	V
$R(T)$	feeder resistance at conductor temperature T	Ω
I_f	prospective fault current	A
I_r	rated continuous current	A
k	instantaneous trip multiple	-
$I_{inst} = k I_r$	instantaneous threshold	A
K	I^2t rating — the let-through energy	A^2s
t_{min}	minimum switching time	s
Δ	coordination margin	s

1. Prospective fault current

Bolted fault — zero fault resistance — at the far end of the feeder, with the source treated as ideal, so this is an upper bound.

$$I_f(T) = \frac{V}{R(T)}$$

Expanded through the T01 resistance, with the round-trip length $L_c = 2L$:

$$I_f(T) = \frac{VA}{\rho_{20} [1 + \alpha(T - T_{ref})] \cdot 2L}$$

The temperature ratio is where the finding lives. V , A and L all cancel, so the 6.28x swing belongs to the material, not to this outpost:

$$\frac{I_f(T_1)}{I_f(T_2)} = \frac{1 + \alpha(T_2 - T_{ref})}{1 + \alpha(T_1 - T_{ref})} \Rightarrow \frac{I_f(100K)}{I_f(400K)} = \frac{1.4274}{0.22740} = 6.28$$

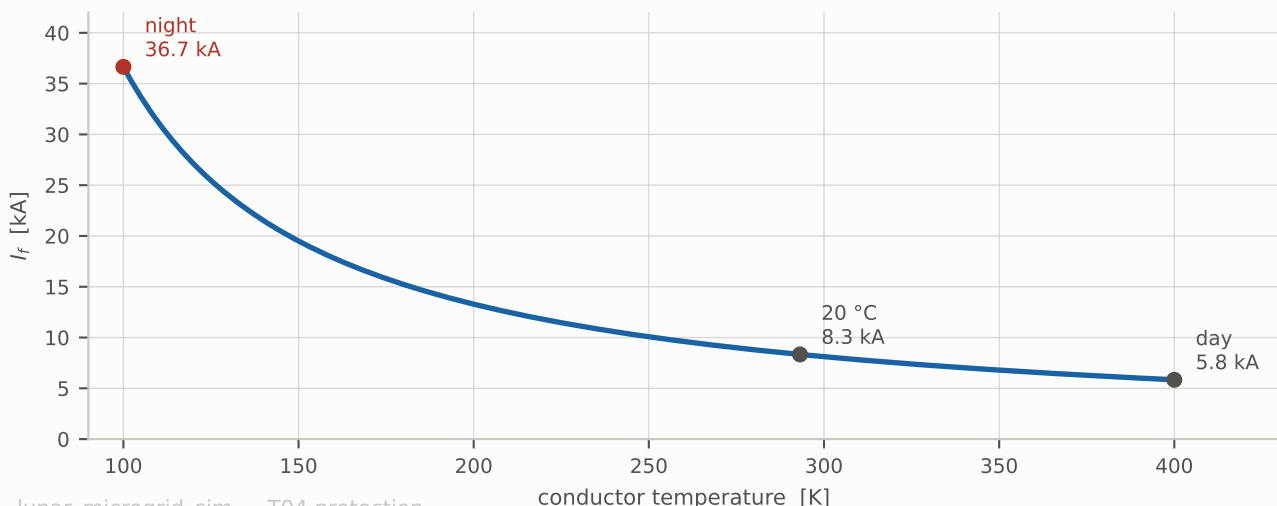
Worked — battery feeder, $A = 36.81 \text{ mm}^2$, $L = 10 \text{ m}$, $V = 120 \text{ V}$, $T = 100 \text{ K}$:

$$R = 6.026\text{e-}9 \times 20 / 36.81\text{e-}6 = 3.274\text{e-}3 \Omega$$

$$I_f = 120 / 3.274\text{e-}3 = 36\ 650 \text{ A}$$

That is 88x the feeder's 417 A rating. No mechanical contact opens 36 kA of direct current and is ever useful again.

Battery feeder. Protection must be designed against the COLD case; losses against the hot one.



2. The trip curve

Three regions, and the order they are tested in is load-bearing.

Piecewise, tested in this order:

$t(I) = \infty$	$I \leq I_r$	never trips
$t(I) = t_{min}$	$I \geq I_{inst} = k I_r$	instantaneous
$t(I) = K/I^2$	otherwise	$I^2 t$ overload

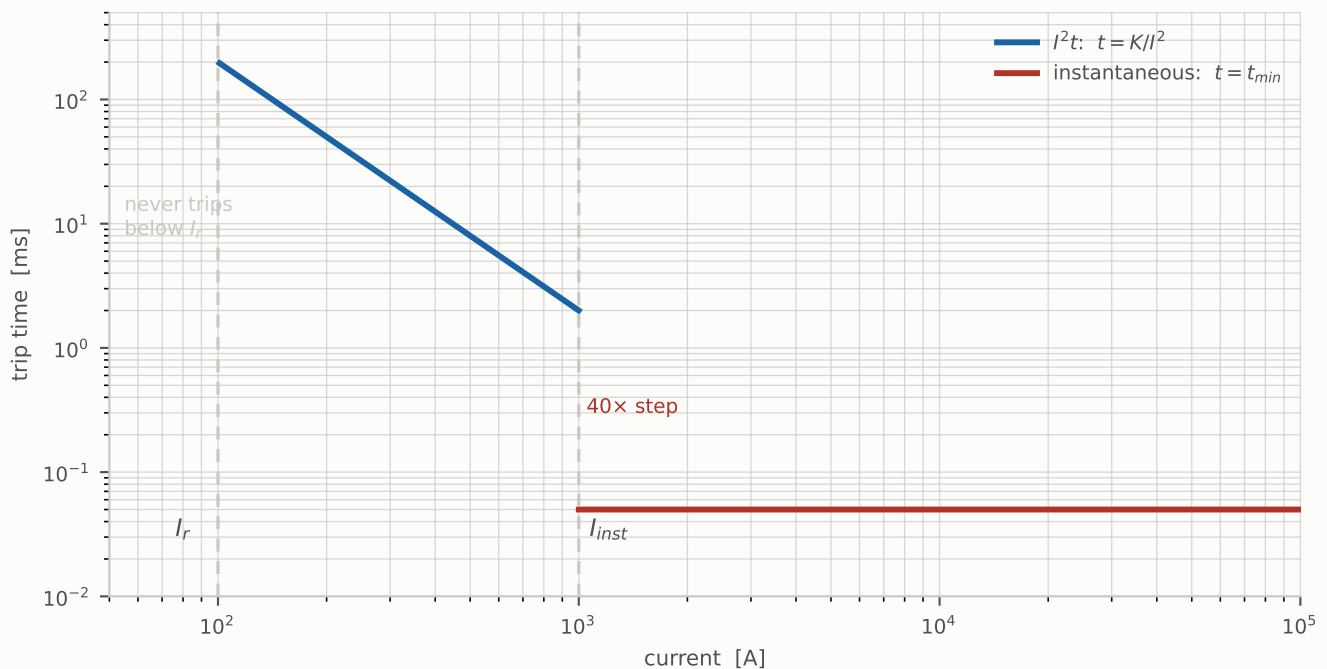
The curve is DISCONTINUOUS at I_{inst} , and deliberately so — the two regions are different physical mechanisms. With the shipped constants:

$$\frac{K}{I_{inst}^2} = \frac{2000}{10^6} = 2 \text{ ms} \quad \rightarrow \quad t_{min} = 50 \mu\text{s}$$

A 40× step down. That is exactly why the branch order matters: test

$I \geq I_{inst}$ BEFORE falling through to K/I^2 , or a dead short is
quoted a trip time the hardware cannot achieve.

A 100 A device. The vertical drop at I_{inst} is the discontinuity, and it is deliberate.



3. Let-through energy

Whether the CABLE survives, as opposed to whether the device noticed.

$$E(I) = I^2 t(I)$$

which by region collapses to:

$$E(I) = \infty \quad I \leq I_r \quad \text{not protecting anything}$$

$$E(I) = K \quad I_r < I < I_{inst} \quad \text{flat — the definition}$$

$$E(I) = I^2 t_{min} \quad I \geq I_{inst} \quad \text{grows as } I^2$$

The middle line is an identity, not a coincidence:

$$I^2 \cdot \frac{K}{I^2} = K$$

which is what makes the flatness test a statement about what an $I^2 t$ device IS, rather than a value copied out of an implementation.

At the worst fault in this outpost:

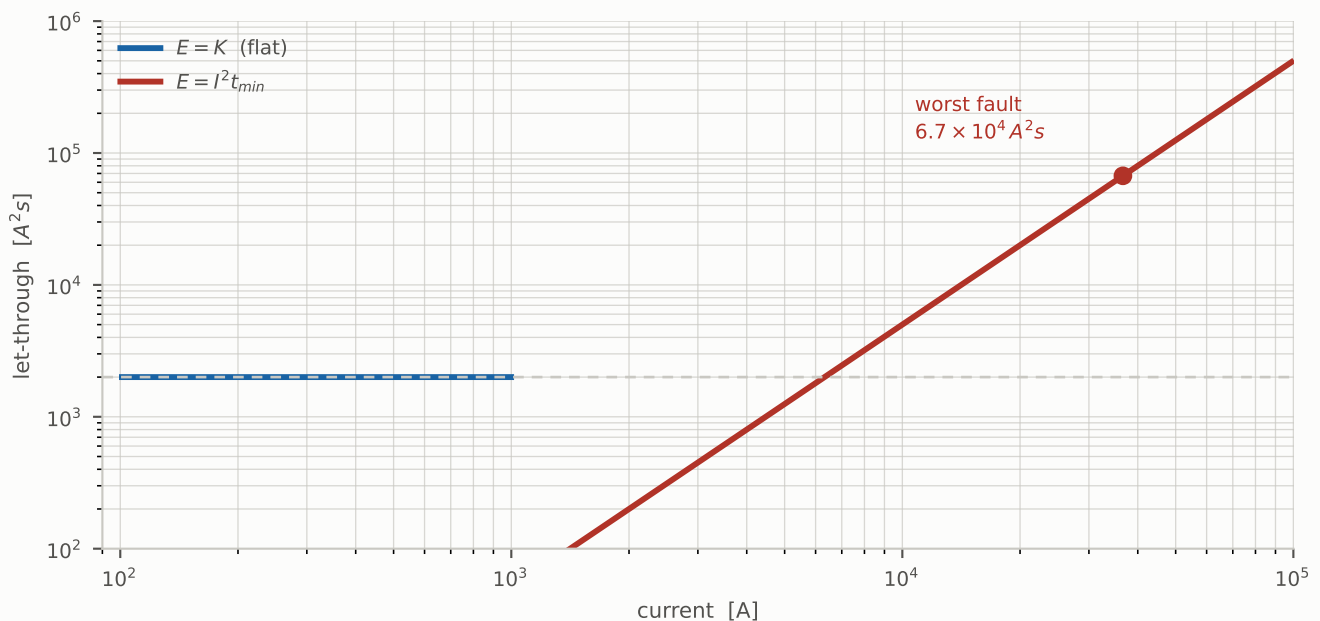
$$E = (36\,650)^2 \times 50 \times 10^{-6} = 6.7 \times 10^4 \text{ A}^2\text{s}$$

against the adiabatic conductor withstand $K_{cable} = k_c^2 A^2$, with k_c from IEC 60364-4-43 (aluminium runs roughly 76–94 with A in mm^2).

At $k_c = 94$ that is about $1.2 \times 10^7 \text{ A}^2\text{s}$ — two orders of margin.

The CABLE is fine. The DEVICE is the constraint.

Flat at K while the device integrates; rising as I^2 once it can only switch as fast as it switches.



4. Selectivity

Zonal protection: the device nearest the fault opens, and nothing else.

$$\text{selective} \iff t_{\text{down}}(I) + \Delta \leq t_{\text{up}}(I)$$

Substitute both devices in their I^2t region and something falls out that is worth more than the formula itself:

$$\begin{aligned} \frac{K_d}{I^2} + \Delta &\leq \frac{K_u}{I^2} \\ \iff I &\leq \sqrt{\frac{K_u - K_d}{\Delta}} \end{aligned}$$

The sting

build_protection gives every device the same K , so $K_u - K_d = 0$ and the bound collapses to $I \leq 0$.

Selectivity in the I^2t region is NEVER achievable with identical I^2t ratings, however far apart the current ratings are.

So with the shipped constants, selectivity survives in exactly one band:

$$I_{r,\text{down}} < I \leq I_{r,\text{up}}$$

where the downstream device is tripping and the upstream one still reads ∞ , because the fault is below ITS rating. Above that band both are in I^2t with equal K (equal times), or both are instantaneous (equal t_{min}). Neither is selective.

That is a real design limitation, not a defect — and it is why practice staggers the I^2t ratings and not merely the current ratings.

Check before you run

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upstream 400 A / downstream 50 A, I = 500 A
  t_up    = 2000 / 5002 = 8 ms
  t_down = t_min      = 50 μs      -> SELECTIVE

upstream 100 A / downstream 100 A, I = 500 A
  both    = 8 ms                      -> not selective

both instantaneous, I = 100 kA
  both    = 50 μs                      -> not selective
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Sources: NASA NTRS 20250000763 (zonal protection, ISS RPCM 4.7 kg vs AMPS module 0.5 kg); IEC 60364-4-43 for the adiabatic withstand constant. Regenerate with docs/make_protection_formulas.py.